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## PAPER

## Theoretical study of the strong piezo-phototronic effect in 2D monochalcogenides for multi-junction solar cells

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E-mail: [mgyan173@gmail.com](mailto:mgyan173@gmail.com) and [vicantwi@yahoo.co.uk](mailto:vicantwi@yahoo.co.uk)**Keywords:** piezophototronic effect, band gap engineering, S-Q limit, multijunction solar cell, voltage mismatch, 2D monochalcogenides materialsSupplementary material for this article is available [online](#)

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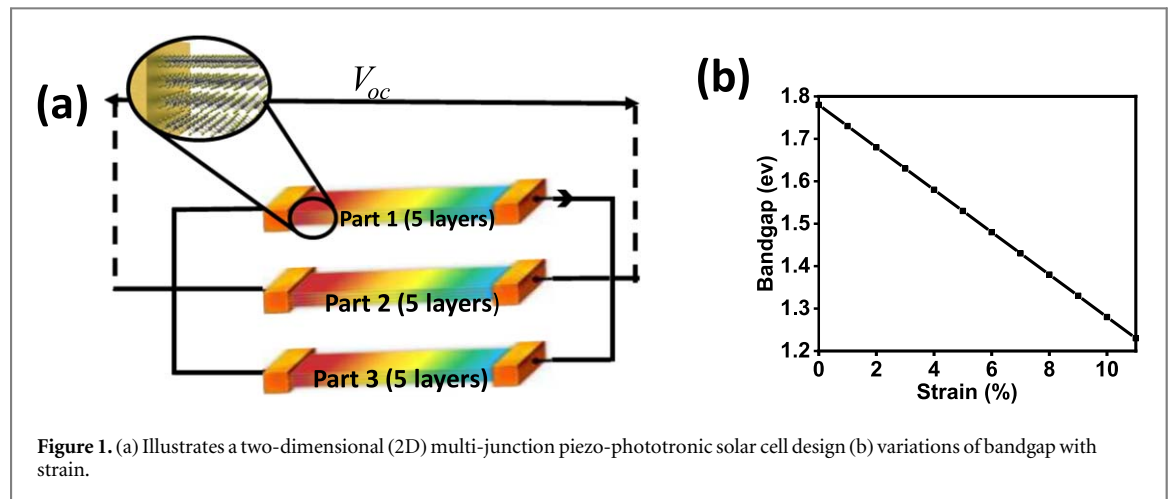
**Abstract**

The piezophototronic effect is a new scientific area that investigates the synergistic interactions of piezoelectric, semiconductor, and photoexcitation features. This effect is seen in crystals lacking inversion symmetry, where applied strain alters electronic transport and provides a way to modify material properties. Monolayer 2D semiconductors, such as transition metal dichalcogenides (TMDCs) and group IV monochalcogenides, have higher piezoelectric coefficients than conventional piezoelectric materials. This study proposes the development of a stable, high-performance multijunction solar cell (MJSC) leveraging the piezo-phototronic effect. The emphasis is on single-type 5-layer 2D monochalcogenides (SnS, SnSe, GeS, and GeSe) with the assistance of strain engineering. Surprisingly, the ultrathin parallel-connected solar cell achieves an electric power conversion efficiency of over 31% when tested under blackbody radiation, surpassing the recognized Shockley–Queisser (S-Q) limit. The piezophototronic effect improves solar cell performance while also addressing voltage mismatch issues. This work introduces a novel approach to developing and manufacturing high-efficiency and robust monolayer multijunction photovoltaic solar cells (MJPSC) based on 2D monochalcogenides.

**1. Introduction**

Piezoelectric materials have several applications in various research disciplines, including sensors, actuators, electric field generators, and optoelectronic devices. Among the 2D materials recognized for their piezoelectric properties are hexagonal boron nitride (h-BN) [1, 2], transition metal dichalcogenides (TMDCs) [3], and group IV monochalcogenides [4]. The recent discovery of nano belts and nanowires has given rise to a new category of devices: nano piezoelectric devices, which combine the properties of piezoelectricity with semiconductor [5, 6]. Examples of these technologies are Field Effect Transistors (FETs) [7, 8], Nano-generators [9, 10], solar cells [11, 12], and piezoelectric diodes [13].

This groundbreaking combination of nanotechnology and piezoelectricity opens up new possibilities for manufacturing tiny and efficient electronic devices with a wide range of capabilities [14, 15]. These semiconductor materials, which include GeS, GeSe, SnS, and SnSe, have structural similarities to black phosphorus and distinct orthorhombic topologies. Their widespread use in a variety of fields, including optoelectronics, piezoelectrics [16], photodetectors, sensors [17, 18], and thermoelectrics, is motivated by their favourable properties, which include low toxicity, environmental friendliness, chemical stability, an excellent temperature range, and abundance in the earth's crust [19, 20]. In their monolayer form, these 2D group IV monochalcogenides are kept together by weak Van der Waals contacts, resulting in a larger band gap than in their bulk state [21, 22]. Defective GeSe, in particular, stands out as a viable alternative for use in photodetectors



and solar cells, with having a strong optical absorptance of 38%, strain tolerance of up to 27%, and a direct band gap [23, 24]. Furthermore, monolayer group IV monochalcogenides have high piezoelectricity, which finds uses in a variety of fields [25, 26]. These 2D materials, notably group IV monochalcogenides, have received a lot of attention in material chemistry, nanotechnology, and condensed matter physics because of their potential and extraordinary features [27].

In response to energy shortages and concerns about environmental contamination, significant efforts have been made to harness renewable and green energy sources. Among the many technologies, solar cells stand out as a viable answer to global energy concerns [28]. Multijunction solar cells, which are made up of various p–n junction semiconducting sub-cells organized in decreasing band gap energies from top to bottom, offer a way to increase solar cell efficiency [29]. Innovations such as concentrated solar systems [30] and six-junction solar cells [31] have accelerated solar cell research. Current research into 2D materials shows that strain can efficiently regulate the band gap [32, 33], with an astonishing in-plane strain limit of up to 27% [34]. Furthermore, strain-induced piezoelectric charges can significantly lower the Schottky barrier of metal-semiconductor interfaces, thereby reducing thermoelectric voltage loss and enhancing solar cell performance [35]. A multi-junction piezo-phototronic solar cell can be developed using a monolayer material by applying varying strains to its sub-cells. This approach offers promising applications due to its high conversion efficiency and straightforward technical requirements.

While several studies have investigated the performance and improvement of multijunction solar cells during temperature changes, there is a significant gap in the existing study addressing 2D single-type monochalcogenides, to the author's knowledge. This study aims to fill this gap by evaluating the performance of a multijunction solar cell that leverages the piezophototronic effect and utilizes a five-layer, triple-junction, 2D monolayer structure composed of group IV monochalcogenides.

This work presents an innovative design for piezo-phototronic multi-junction solar cells that use a single type of two-dimensional material. Figure 1(a) depicts the construction of a piezophototronic multijunction solar cell, which consists of three distinct sub-cells: the top cell (Part 1), the middle cell (Part 2), and the bottom cell (Part 3) with each having five stacked monolayer GeSe. Various strains are applied to the top, middle, and bottom cells, respectively. The GeSe stacking layers act as the light-absorbing material sandwiched between two contacts. One contact operates as a Schottky barrier, separating photon-generated carriers, while the other acts as an ohmic contact, allowing current to flow without rectification. Figure 1(b) shows the modulation of bandgap with different strains, demonstrating how applied strain affects the bandgap of two-dimensional semiconductors and the formation of piezoelectric charges.

The bandgap of monolayer GeSe decreases considerably from 1.78 eV to 1.12 eV when external strain increases, allowing for the absorption of lower-energy photons. Furthermore, the piezoelectric charges increase significantly with strain, surpassing those of typical piezoelectric semiconductors due to monolayer GeSe's capacity to endure up to 27% strain [36–38].

In electrical circuit theory, the piezo-phototronic multijunction solar cell is analogous to a battery made up of multiple sub-cells linked in parallel, requiring matching open-circuit voltage ( $V_{oc}$ ) among sub-cells to reduce inherent loss. This research shows how the piezo-phototronic effect may minimize  $V_{oc}$  mismatch between sub-cells of various layers while improving the performance of piezo-phototronic multijunction solar cells. As a result, a triple-junction piezo-phototronic multijunction solar cell composed of 15 stacked monolayer GeSe achieves a power conversion efficiency of around 33%, above the Shockley–Queisser limit. This discovery highlights the ability of piezo-phototronic multi-junction solar cells based on single-type two-dimensional

materials to exceed the theoretical Shockley–Queisser limit by leveraging the photo-phototronic effect and deformation potential.

This study may be utilized to develop a stable, reliable, and highly efficient solar cell.

### 1.1. The basic theory of multijunction solar cells

#### 1.1.1. A 2D multijunction solar cell's theoretical analysis employing a detailed balancing limit

Treating the illumination as a blackbody with  $T_s = 6000$  K, the spectral photon flux can be derived from Planck's law as follows:[39, 40]

$$F_s = f_\omega \frac{2\pi}{h^3 c^2} \frac{E^2}{e^{E/kT_s} - 1} \quad (1)$$

where, respectively, the  $c$ ,  $k$ ,  $h$ ,  $E$  and  $f_\omega$  represent the speed of light, Boltzmann's constant, and Planck's constant, the energy of photons and geometry factor respectively, and  $f_\omega = 2.18 \times 10^{-5}$  [41]. Moreover, a solar cell's current density, which is produced by absorbing the incident photons, is determined by

$$J_{sc} = q \int_0^\infty a(E) F_s dE = q f_\omega \int_0^\infty a(E) \frac{2\pi}{h^3 c^2} \frac{E^2}{e^{E/kT_s} - 1} dE \quad (2)$$

In the equation above, the terms  $a(E)$  refers to the rate at which photons with energy  $E$  are absorbed and  $q$  is the electron charge. Even when not exposed to illumination, the solar cell emits and absorbs light equivalent to that of a blackbody at  $T_c = 300$  K, as it is thermodynamically balanced at this high ambient temperature. However, when illuminated, the solar cell may not remain thermodynamically balanced with its surroundings. According to previous research [42], the reverse saturation current ( $J_0$ ) is expressed as

$$J_0 = f_g \frac{2\pi q}{h^3 c^2} \int_0^\infty e(E) E^2 e^{-E/kT_c} dE \quad (3)$$

where  $e(E)$  is the probability of photon emission, at thermal equilibrium, the current densities must be balanced  $e(E) = a(E)$ ,  $f_g$  is a geometry factor. The geometry factor ( $f_g$ ) = 2 [43].

The open-circuit voltage  $V_{oc}$  can be expressed as [35, 44]

$$V_{oc} = \frac{kT_c}{q} \left[ \ln \left( \frac{J_{sc}}{J_0} \right) + 1 \right] \quad (4)$$

In this study, we assume  $a(E) = 0$  for photons with energy  $E < E_g$ . Additionally, light absorption can reach up to 20% for photon energies exceeding the bandgap. As a result,  $a(E) = 50\%$  for each sub-cell which consists of five monolayer materials whose photon energy exceeded the energy band gap. The configuration of a parallel triple-junction solar cell used is indicated in figure 1(a). This work made use of parallel connection since piezo charges created as a result of the application of strain help match the open circuit voltage. The total current density is equal to the sum of the current density of the subcell is expressed as

$$\begin{cases} J_{sc}^1 = q f_\omega \int_{E_g^1}^\infty a(E) \frac{2\pi}{h^3 c^2} \frac{E^2}{e^{E/kT_s} - 1} dE \\ J_{sc}^2 = q f_\omega \int_{E_g^2}^\infty a(E) \frac{2\pi}{h^3 c^2} \frac{E^2}{e^{E/kT_s} - 1} dE - a(E) J_{sc}^1 \\ J_{sc}^3 = q f_\omega \int_{E_g^3}^\infty a(E) \frac{2\pi}{h^3 c^2} \frac{E^2}{e^{E/kT_s} - 1} dE - a(E) J_{sc}^2 - a(E) J_{sc}^1 \end{cases} \quad (5)$$

Where  $J_{sc}^1$ ,  $J_{sc}^2$  and  $J_{sc}^3$  are the current density for the top cell, middle cell, and bottom cell, respectively.

From Kirchhoff's law, the summation of the total current density for all sub-cells represents the overall current density of multijunction solar cells. A parallel solar cell's configuration (J-V) characteristic is defined by

$$\begin{cases} J(V) = \sum_{i=1}^{2,3} \left( J_0^i \left[ \exp \left( \frac{qV}{kT_c} \right) - 1 \right] - J_{sc}^i \right) \\ J_{sc}^p = J(V=0), \quad J(V_{oc}^0) = 0 \end{cases} \quad (6)$$

Where  $i$  is the number of the sub-cell,  $J_0^i$ ,  $J_{sc}^p$ ,  $V_{oc}^0$ , and  $J_{sc}^i$  are saturation current density, total short circuit current density, open-circuit voltage and short circuit density of every sub-cell, respectively.

This work considers stacked multi-monolayer Monochalcogenides material. When a piezoelectric semiconductor material is strained, piezoelectric charges are created on both sides of the material, causing the band gap to change. the stresses applied to Parts 1, 2, and 3 are referred to as  $s_{11}^1$ ,  $s_{11}^2$  and  $s_{11}^3$  respectively. The bandgaps for the top, middle, and bottom cells are determined by [45]:

**Table 1.** Shows dielectric constant, deformation potential, and piezoelectric constant of selected 2D monochalcogenides piezoelectric semiconductor materials.

| Material(s)                  | SnS         | SnSe       | GeS         | GeSe        |
|------------------------------|-------------|------------|-------------|-------------|
| $E_s$ (meV)                  | 116.00 [46] | 87.50 [47] | 160.00 [48] | 145.00 [49] |
| $e_{11}$ (C/m <sup>2</sup> ) | 4.11 [25]   | 4.26 [25]  | 1.36 [25]   | 2.37 [25]   |
| Bandgap(ev)                  | 1.86 [50]   | 1.62 [51]  | 1.77 [51]   | 1.78 [51]   |
| $\epsilon_s$                 | 15.78 [52]  | 17.95 [52] | 11.75 [53]  | 10.10 [53]  |

$$\begin{cases} E_g^1 = E_g^0 - 100 \times E_s s_{11}^1 \\ E_g^2 = E_g^0 - 100 \times E_s s_{11}^2 \\ E_g^3 = E_g^0 - 100 \times E_s s_{11}^3 \end{cases} \quad (7)$$

where  $E_g^0$  and  $E_s$  is the material bandgap without strain and with a variation of strains respectively.

The J-V characteristic of the piezo-phototronic multijunction (PMJSC) solar cell is obtained using

$$\begin{cases} J(V) = \sum_{i=1}^{2,3} \left( J_0^i \exp \left( -\frac{q e_{11} s_{11}^i W_{piezo}}{2 \epsilon k T_c} \right) \left[ \exp \left( \frac{qV}{kT_c} \right) - 1 \right] - J_{sc}^i \right) \\ J_{sc}^p = J(V=0), V_{oc} = V_{oc}^0 + \frac{P_z W_{piezo}}{2 \epsilon} \end{cases} \quad (8)$$

where  $V_{oc}^0$  and  $V_{oc}$  is the open circuit voltage of MJSC without and with piezoelectric charges,  $e_{11}$ ,  $s_{11}$ ,  $\epsilon$ ,  $W_{piezo}$ ,  $P_z$  are piezoelectric constant, applied strain, material dielectric constant and piezoelectric charge distribution width piezoelectric charges respectively and  $P_z = e_{11} s_{11}$ .

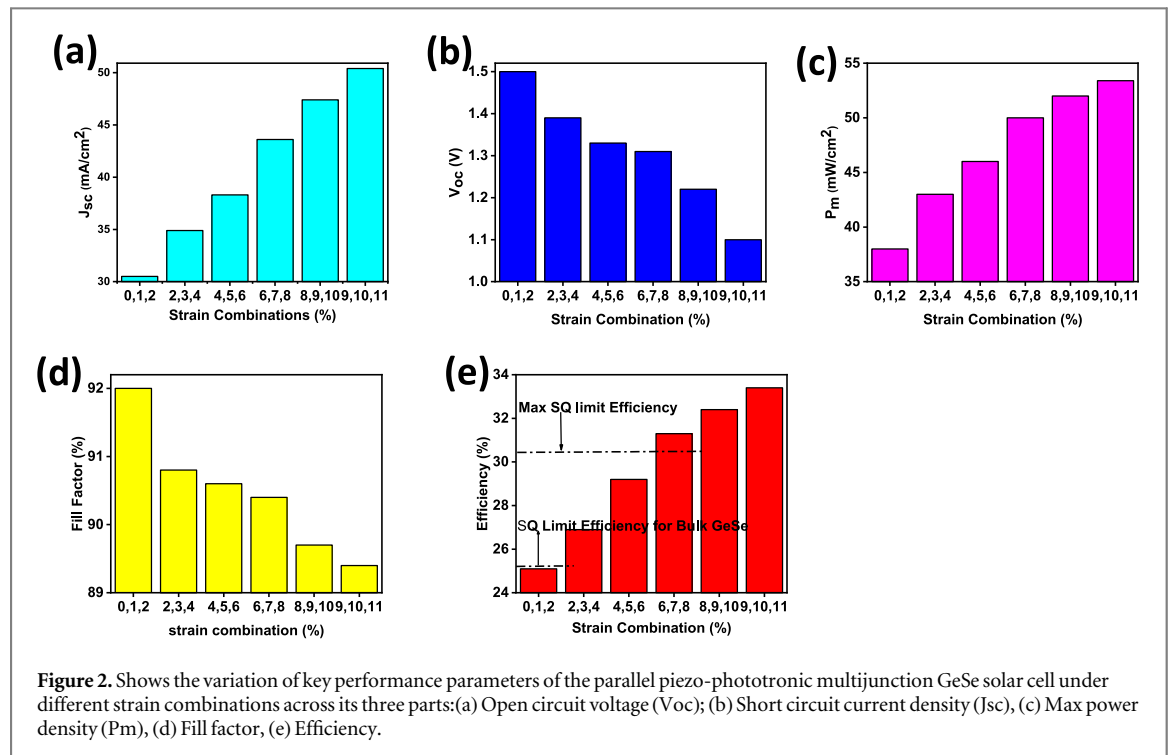
According to the model's equations, the simulated device functions as a photodiode with a metal–semiconductor contact. The asymmetry in carrier direction is driven by a high electric field in the depletion region of the photodiode, which effectively separates photogenerated carriers. A PN junction, characterized by an internal electric field due to carrier depletion, can be utilized in both photodiodes and solar cells. The asymmetry arises from electrodes with differing work functions, which is the principle behind the metal–semiconductor contact. The current–voltage relationship model employs a classical approach using semiconductor drift–diffusion equations, as demonstrated in experiments with monolayer MoS<sub>2</sub> devices. In this work, the Shockley–Queisser efficiency limit of the solar cell, based on monolayer MoS<sub>2</sub>, is described using these semiconductor drift–diffusion equations. The piezo-phototronic effect can be introduced to enhance both the open-circuit voltage and also solve the voltage mismatch associated with a parallel arrangement of the subcells.

The absorption rate represents the probability of absorbing photons with energies above the bandgap ( $E_g$ ). The optical properties of monolayer and few-layer 2D materials can be assessed using three complementary spectroscopic techniques: photoluminescence (PL), optical absorption, and photoconductivity. By analyzing the absorption spectra, PL spectra, and photoconductivity spectra, the absorption rate for mono- and few-layer 2D films can be determined. For simplicity, our model employs an average absorption rate.

## 2. Results and discussion

The power conversion efficiency (PCE) of piezoelectric multi-junction solar cells (PMJSCs) relies on the properties of the materials used. Specifically, 2D materials with high  $e_{11}/\epsilon_s$  ratios can effectively adjust the Schottky barrier height (SBH) through the application strain. Additionally, these solar cells materials require a significant bandgap alteration of the materials under strain. Consequently, the PCE of PMJSCs is influenced by various characteristics of 2D piezoelectric materials, including the piezoelectric constant, dielectric constant, bandgap, and bandgap deformation potential. Table 1 presents the parameters of 2D piezoelectric semiconductor materials used for calculating the PCE of PMJSCs.

Previous studies have shown that monolayer GeSe can resist stresses greater than 27% before breaking [54]. Monolayer monochalcogenides, as well as a few layers, have been proven in studies to be flexible up to the maximum strain limit [39, 43]. Various piezoelectric materials have stable piezoelectric constants over a wide range of stresses [55]. Similarly, the five monolayer monochalcogenide materials' piezoelectric constant is projected to stay constant. Furthermore, because each monolayer in the five-layer structure is oriented in the same polarization direction, the piezoelectric constant is expected to be equal to that of the monolayer [56]. The piezoelectric charges distribution width  $W_{piezo}$  is 0.25 nm [37], and the material dielectric constant for single

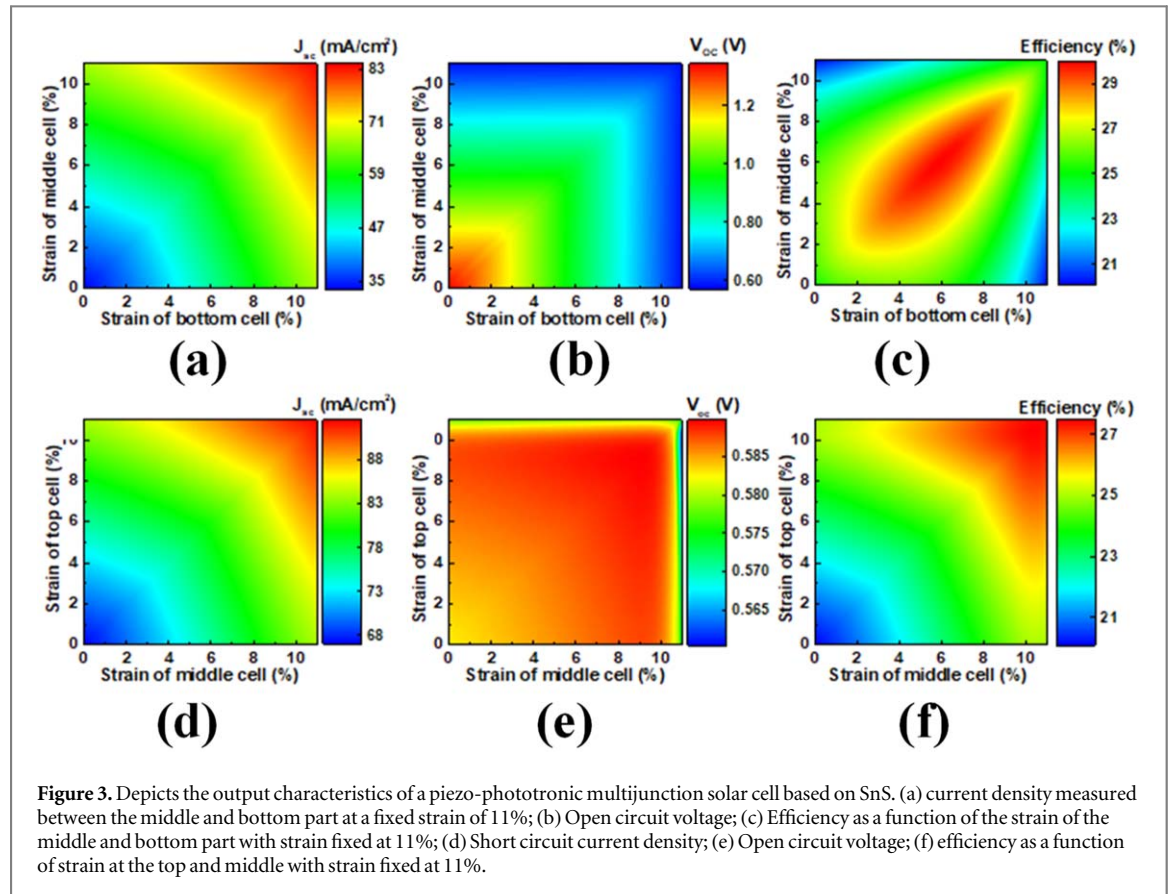


layer SnS, SnSe, GeS, and GeSe are provided as 15.78, 17.95 [52], 11.75 and 10.1 [53], single-layer SnS, SnSe, GeS, and GeSe have respective band-gap energies of 1.86 eV, 1.62 eV, 1.77 eV, and 1.78 eV [50, 51].

Figure 2(a) shows how the short circuit current density ( $J_{sc}$ ) varies with different strain combinations. As the strain combination increases, the bandgap reduces, resulting in increased photon absorption and, thus, a rise in  $J_{sc}$ . Figure 2(b) illustrates the decrease in open circuit voltage ( $V_{oc}$ ) with increasing applied strain.  $V_{oc}$  is governed by the material bandgap, and since strain affects the bandgap,  $V_{oc}$  is slightly lower than the bandgap due to voltage loss. Furthermore, as GeSe is a piezoelectric semiconductor material, strain-induced piezoelectric charges can further reduce voltage loss. Figure 2(c) depicts the optimum power density of a piezo-phototronic multijunction solar cell under various strain combinations. Photon absorption improves as strain increases, and strain-induced piezoelectric charges reduce voltage loss, resulting in a higher maximum power density. Figure 2(d) depicts an in-depth analysis of the fill factor (FF) of a piezo-phototronic multijunction solar cell under various strain combinations. The graph shows that when the applied strain combination advances, the FF decreases steadily, which is consistent with theoretical assumptions for single-junction solar cells. Furthermore, figure 2(e) offers useful information about the efficiency of the piezo-phototronic multijunction solar cell under various strain combinations. Surprisingly, increasing the applied strain combination improves the solar cell's efficiency significantly. Notably, the efficiency exceeds the Shockley–Queisser limit efficiency associated with a bandgap of 1.78 eV (typical of GeSe) when the strain combination across the three sub-cells remains around 2%, 3%, and 4%.

Furthermore, the efficiency of the piezo-phototronic multijunction solar cell has the potential to exceed the maximum Shockley–Queisser limit efficiency, which is usually recorded at a bandgap of 1.3 eV. This astonishing result occurs when the strain combinations in the three sub-cells are kept at 6%, 7%, and 8%. Figures 2(d) and (e) shows that strain has a considerable impact on both the fill factor and the efficiency of piezo-phototronic multijunction solar cells. Figure 2(e) illustrates how the maximum PCE of the parallel PMJSC varies with the photon strain combination. The maximum PCE occurs when the bottom cell experiences a strain of 1%–3%. This is attributed to strain reorganization and the optimization of thermalization loss through strain. The fill factors (FF) corresponding to figure 2 are presented in Figures S1. These findings show the possibility of exceeding the recognized efficiency limits by strategically manipulating strain combinations across sub-cells.

We investigate the output characteristics of piezo-phototronic multi-junction solar cells (PMJSCs) using 2D SnS. Figure 3(a) depicts the impact of strain on the middle and bottom cells, while the top cell's current density ( $J_{sc}$ ) remains unchanged. As more photons are absorbed, the band gap narrows and current density increases. Furthermore, figure 3(b) shows that when strain increases on the middle and bottom cells, the open-circuit voltage ( $V_{oc}$ ) decreases, although the top cell stays unaltered. This arises because  $V_{oc}$  is reliant on the bandgap, and a strain-induced drop in the bandgap leads to a lower  $V_{oc}$ . Figure 3(c) depicts the efficiency of PMJSC using 2D SnS material, with the top cell at 0% strain and the bottom and middle cells stretched to 11%. The decrease in



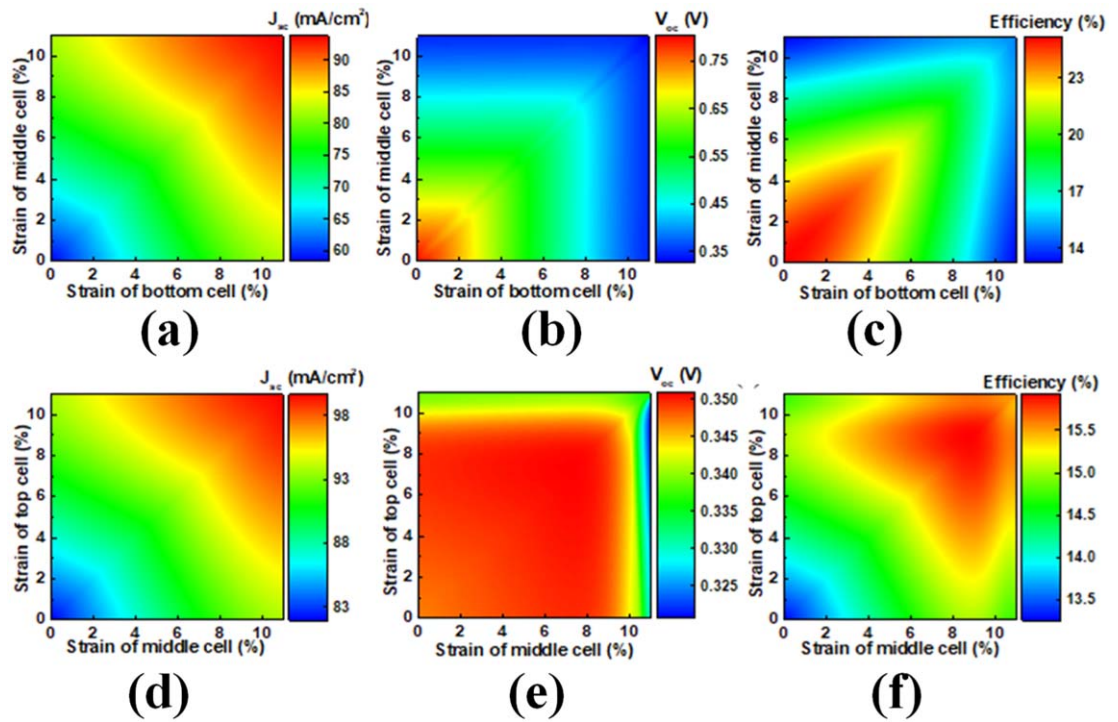
**Figure 3.** Depicts the output characteristics of a piezo-phototronic multijunction solar cell based on SnS. (a) current density measured between the middle and bottom part at a fixed strain of 11%; (b) Open circuit voltage; (c) Efficiency as a function of the strain of the middle and bottom part with strain fixed at 11%; (d) Short circuit current density; (e) Open circuit voltage; (f) efficiency as a function of strain at the top and middle with strain fixed at 11%.

bandgap facilitates voltage matching and enhances energy output, leading to a peak efficiency exceeding 31%. Figures 3(d) and (e) show the effect of strain on the open-circuit voltage and current density of the top and middle cells, respectively, when 11% strain is applied to the bottom cell, which supports the findings from figures 3(a) and (b). The PCE of parallel multijunction solar cells (PMJSCs) can vary based on the strain changes in the 2D material layers used as top and middle cells. The maximum PCE of these PMJSCs can reach up to 31.0%, as illustrated in figure 3(c). As depicted in figure 3(d), when different fixed strains are applied to the 2D material in the middle cell, the bandgap energy ( $E_g$ ) changes, affecting the PCE. The PCE of a parallel PMJSC can fluctuate with the strain variations in the bottom cell and its absorption rate, a trend comparable to what is shown in figure 4(c) and 5c. The highest PCE initially increases and then decreases as the strain in the top/bottom subcell layers changes.

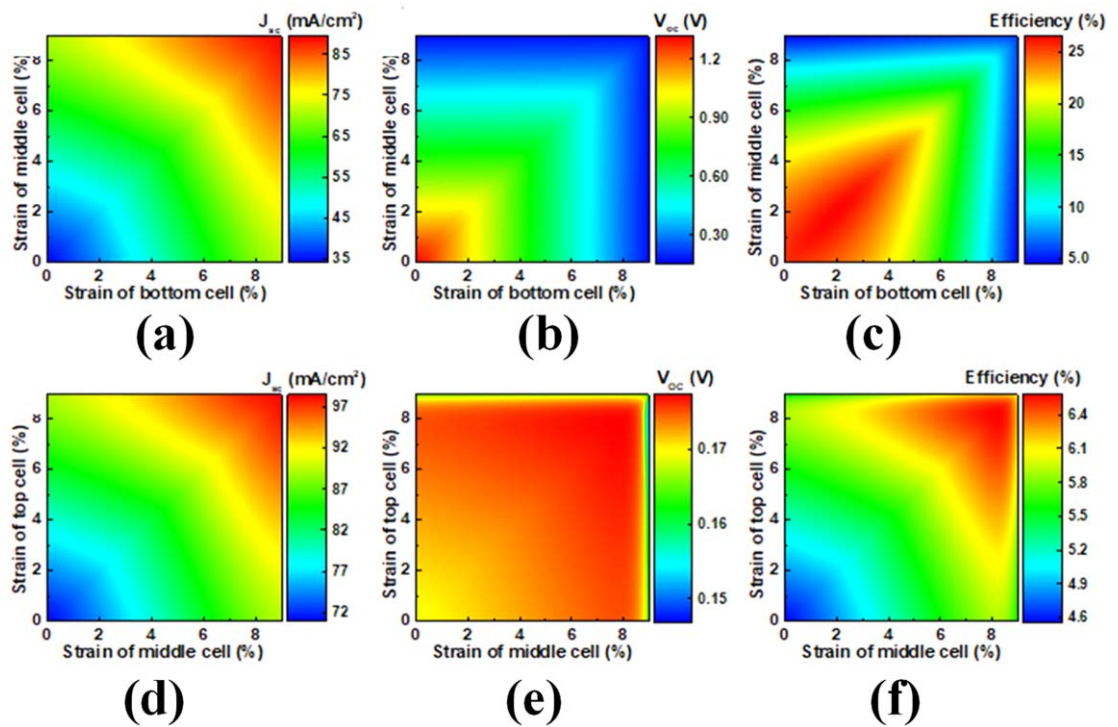
The efficiency of 2D SnS PMJSC fluctuates with the strain of the top and middle cells, with the bottom strain set to 11%, as illustrated in figure 3(f). Improved voltage matching with greater strain on the top and bottom cells adds to higher efficiency.

Similarly, figure 4 depicts PMJSC output characteristics using 2D SnSe. Figure 4(a) shows how strain affects short-circuit current density ( $J_{sc}$ ) in the middle and bottom cells, but not in the top cell. Figure 4(b) shows a reduction in open-circuit voltage ( $V_{oc}$ ) as strain increases on the middle and bottom cells, while the top cell remains unstrained. Furthermore, figure 4(c) shows how PMJSC efficiency varies with strain in the middle and bottom cells, with the top cell strain fixed to zero. The highest efficiency exceeds 24 percent. Figures 4(d) and (e) show the strain-dependent behaviour of current density and open-circuit voltage for the top and middle cells, respectively, with 11% strain applied to the bottom cell, which is similar to figures 4(a) and (b). Figure 4(f) demonstrates how the PMJSC's efficiency varies depending on the strains of the top and middle cells, while the bottom strain is 11% and reaches a maximum value at a given strain combination.

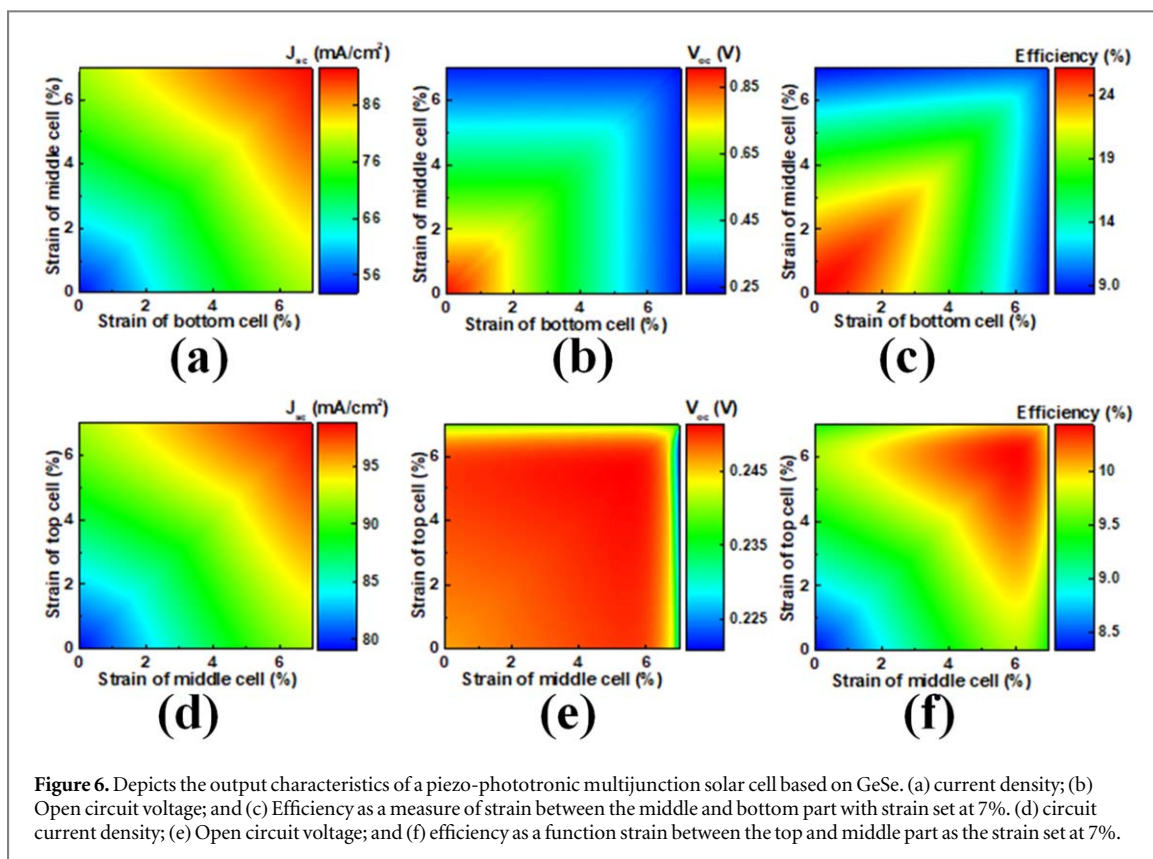
Figures 5(a) and (b) show the performance of PMJSC with 2D GeS. Notably, the top cell stays unstrained, while the middle and bottom cells' short-circuit current density ( $J_{sc}$ ) and open-circuit voltage ( $V_{oc}$ ) alter in response to strain. The connection between  $J_{sc}$ ,  $V_{oc}$ , and strain in the middle and bottom parts is antagonistic, consistent with the Shockley–Queisser theory. Figure 5(c) also shows how strain affects PMJSC efficiency in the middle and bottom portions, but not in the top cell. Notably, efficiency peaks at around 25% for a certain low-strain combination. Figures 5(d) and (e) show the fluctuation in current density and open-circuit voltage with strain at the top and centre parts, which are set at 9%, while the bottom section is unstrained. These results are consistent with those displayed in figures 5(a) and (b). Furthermore, figure 5(f) shows how PMJSC efficiency



**Figure 4.** Output characteristics of a piezo-phototronic multi-junction solar cell based on SnSe. (a) current density; (b) Open circuit voltage; (c) Efficiency as a function of strain between the middle and bottom part with a fixed strain of 11%; (d) Short circuit current density; (e) Open circuit voltage; (f) Efficiency as a function of strain between the top and middle part with a fixed strain 11%.



**Figure 5.** Depicts the output characteristics of a piezo-phototronic multijunction solar cell based on GeS. (a) circuit current density; (b) Open-circuit voltage; and (c) Efficiency as a function of strain between the middle and bottom part with the strain kept at 9%. (d) current density; (e) Open circuit voltage; and (f) efficiency as a function of strain between the top and middle part with strain kept at 9%.



varies with the strain of the top and middle cells, with the bottom section stressed to 9%. Although efficiency peaks at a given strain combination, it remains just below 7%.

The results demonstrate the performance of 2D GeSe-based output in PMJSCs. Under no applied stresses to the top cell, both current density ( $J_{sc}$ ) and open circuit voltage ( $V_{oc}$ ) fluctuate in the middle and bottom regions, as shown in figures 6(a) and (b). Furthermore, figure 6(c) depicts the fluctuation in PMJSC efficiency with strain in the middle and bottom parts when strain is not present. With a certain strain combination, efficiency reaches a high of around 25%. Figure 6(d) and (e) show the variation in current density and open circuit voltage with a constant 7% strain on the top and middle parts, while the bottom section stays strain-free. The interdependent interactions between several PMJSCs that use 2D piezoelectric semiconductors and different combinations of applied strain show significant variety. In this work, strain can obviously vary bandgap of material, and 2D monochalcogenide materials at different strains can be used as sub-cells of piezo-phototronic solar cell to absorb photons of different energy. Significantly, the open circuit voltage will decrease while bandgap decrease. However, the drop of open circuit voltage is decreases owing to strain induced piezoelectric potential. This enhance voltage match of parallel piezo-phototronic multijunction solar cell. This is unique advantages of piezoelectric semiconductor. This variety is mostly due to the various material characteristics, which have a substantial impact on performance. Notably, 2D monochalcogenides have excellent piezoelectric characteristics and deformation potential, allowing PMJSCs-based materials like SnS2 to exceed the Shockley–Queisser limit. Under applied strain, the efficiency of PMJSCs using various types of 2D materials can also improve, but not to maximum levels. This improvement is due to the energy emission and voltage matching capabilities of PMJSCs. Essentially, PMJSCs made of a single type of 2D material have the potential to reach the maximum Shockley–Queisser limit, resulting in extremely high efficiency.

This study demonstrates how strains have a significant influence on material band gaps and how 2D materials subjected to various strains may function as sub-cells in piezo-phototronic solar cells to catch photons over a wide range of energies. Narrowing bandgaps reduce open-circuit voltage significantly, but the strain-induced piezoelectric potential offsets this voltage loss, improving the voltage-matching capabilities of parallel piezophototronic multijunction solar cells [25, 41]. The distinct properties of piezoelectric semiconductors are highlighted, particularly their potential for roll-to-roll engineering production, as proven in prior research involving the creation of two-dimensional materials in 100-meter-long rolls [57]. In addition, series multijunction solar cells' conversion efficiency has been studied [58, 59]. Because series arrangement often ends with the problem of current mismatch, this significantly limits the total range of the possible material band gaps. This study made use of a parallel configuration. The voltage mismatch that frequently arises in parallel structures

**Table 2.** Comparison of efficiency and normalized efficiency per thickness (nm) between typical output performance of 2D MJPSCs and mainstream solar cells.

| Material(s)        | Efficiency(%) | Normalized efficiency(%) |
|--------------------|---------------|--------------------------|
| 2D Perovskite [60] | 12.53         | 0.054                    |
| InP [61]           | 22.1          | 0.006                    |
| Si                 | 20.6          | 0.001                    |
| GaAs [62]          | 29            | 0.029                    |
| GaAs/GaInP [63]    | 35.36         | 0.005                    |
| DASH [64]          | 35.36         | 0.093                    |
| 2D MJPSC**         | 31            | 1.840                    |

can be corrected by strain-induced piezoelectric potential [35], which allows the piezophototronic multijunction solar cell to have a wider range of alternative materials and fewer difficulties in engineering design.

In addition, the typical output performance of 2D MJPSCs and popular mainstream solar cells based on both single-junction and multi-junctions including 2D Perovskite [60], InP [61], Si, GaAs [62], GaAs/GaInP [63], and Dopant-Free Asymmetric Hetero-contact Silicon(DASHS) [64] are compared in table 2. For traditional multi-junction solar cells, power density is extremely limited due to the thicker stacking structure and complex manufacturing process. These highly improved efficiencies come at the expense of power density. For 2D MJPSCs, the active layers are of the thinnest at only 5–10 nm, and the total power conversion efficiency is comparably high at over 31%, which results in the highest normalized efficiency of over 1.8% when blackbody spectrum is employed. Such high-normalized efficiency per thickness in nanometer is due to its ultra-thin nature of a monolayer 2D TMDCs. Its high-performance design enables next-generation low-cost, high-efficiency solar cells.

### 3. Conclusion

Finally, we presented a ‘multi-junction solar’ cell using monolayer 2D piezoelectric materials and studied its output characteristics and efficiency. Strain alters the bandgap and generates piezoelectric charges, which influence the open-circuit voltage and short current density significantly. The triple-junction piezo-phototronic multijunction solar cell has an energy conversion efficiency of more than 31% when sunlight is considered to be a black body at 6000K. Notably, the efficiency of this device exceeds the Shockley-Queisser limit for bulk monochalcogenides at various strain combinations, indicating that it has prospective applications. This discovery lays the path for the creation of extremely efficient solar cells that combine the piezophototronic effect with strain management.

### Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

### Conflict of interest

The authors declare that they have no conflict of interest.

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